Known Values

$V_{IN} = 12V$

$V_{out} = 5V$

$I_{out} = 2A$

$f_{sw} = 500kHz$

$V_{ripple} = 1\text{-}5\% = 50mV$ (2.1)

Inductor Ripple (LIR) = 0.3 Pg.16

$V_{ref} = 0.8V$

$I_{ss} = 2\mu A$

$\Delta I_L = 0.583A$

$2 \leq t_{ss} \leq 10$ ms $\therefore t_{ss} = 2ms$

Solved

$C_{boot} = 0.1\mu F$

$L = 9.72\mu H$

$C_o = 80\mu F$

$R_{osc} = 49.9k\Omega$

$C_{ss} = 5nF$

$C_{in} = 20\mu F$

Values to find

~~L~~, ~~C_o~~, C_{in}, ~~C_{boot}~~, ~~R_{osc}~~, ~~C_{ss}~~, R_c, C_c

C_{boot} given Pg. 3 : 0.2 µF ceramic

$$L = \frac{V_{IN} - V_{out}}{I_o \cdot LIR} \cdot \frac{V_{out}}{V_{IN} \cdot f_{sw}} = \frac{12-5}{2 \cdot 0.3} \cdot \frac{5}{12 \cdot 500kHz} \approx 9.72\mu H$$

$$\Delta i_L = \frac{V_{in} - V_{out}}{L} \cdot \frac{V_{out}}{V_{in} f_{sw}} = \frac{12-5}{10\mu H} \cdot \frac{5}{12 \cdot 500kHz} = 0.583$$

$$C_o > \frac{\Delta I_{ripp}^2 \cdot L}{2 \cdot V_{out} \cdot V_{ripple}} = \frac{2^2 (10\mu H)}{2 \cdot 5 \cdot 0.05} = 80\mu F$$

$$R_{osc} = 21.82 \cdot f_{sw}^{-1.167} = 21.82 \cdot 500kHz^{-1.167} = 21.82 \cdot 0.5MHz^{-1.167} \approx 49.9k\Omega$$

$$t_{ss}(ms) = \frac{C_{ss}(nF) \cdot V_{ref}}{I_{ss}(\mu A)} \Rightarrow \frac{t_{ss} \cdot I_{ss}}{V_{ref}} = C_{ss} \Rightarrow \frac{2 \cdot 2}{0.8} = \frac{4}{0.8} = 5nF$$

$$50mV = \frac{2A \cdot 0.25}{C_{IN} \cdot 500000} \Rightarrow C_{IN} = \frac{2A \cdot 0.25}{50mV \cdot 500000} = 20\mu F$$